Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

Ace editor not ready. Perhaps reload page?

Falling back to raw text area.

```
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int i = 1;
    int s = 1;
    int count = 2;

    while (1) {
        count++;

        if (s > n)
            break;

        i++;
        count++;
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main()
{
    int n;
    scanf("%d", &n);

    int count = 0;

    if (n == 1)
    {
        count++;
    }
    else
    {
        count++;

        for (int i = 1; i <= n; i++)
        {
            count++;

            for (int j = 1; j <= n; j++)
            {
                count++;
                count++;
                count++;
                break;
            }
```

```
29
30              count++;
31          }
32
33          count++;
34      }
35
36      printf("%d", count);
37
38      return 0;
39  }
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>
int main(){
    int num;
    scanf("%d",&num);
    int counter=0;
    for(int i=1;i<=num;i++){
        counter++;
        if(num%i==0)
        {
            counter++;
        }
        counter++;
    }
    counter--;
    printf("%d",counter+2);
    return 0;
}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include <stdio.h>
void function(int n,int *c1){
    int c= 0;
    (*c1)++;
    for(int i=n/2; i<n; i++){
        (*c1)++;
        for(int j=1; j<n; j = 2 * j){
            (*c1)++;
            for(int k=1; k<n; k = k * 2){
                (*c1)++;
                c++;
                (*c1)++;
            }
            (*c1)++;
        }
        (*c1)++;
    }
    (*c1)++;
}
int main(){
    int n,c1=0;
    scanf("%d",&n);
    function(n,&c1);
    printf("%d",c1);
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

void reverse(int n)
{
    int rev = 0, remainder;
    int count = 0;
    int digits = 0;
    int temp = n;

    while(temp != 0)
    {
        digits++;
        temp /= 10;
    }

    count = 4 * digits + 3;

    while(n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n /= 10;
    }

    printf("%d\n", count);
}

int main()
{
    int n;
    scanf("%d", &n);
    reverse(n);
    return 0;
}
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course